KAIZEN MATHS

Lesson starter

Algebraic Simplification

Like and Unlike Terms

Name _____

S1

Simplify:

$-x - 6x - 8x$

S2

Simplify:

$-3x - 3y + 2x - 8$

S3

Simplify:

$3x + 3 + 2x$

S4

Simplify:

$4x + 3x + 2x$

Show your working

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ALG-3B24-L1 · S

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Exit ticket

Algebraic Simplification

Like and Unlike Terms

Name _____

E1

Simplify:

$-x - 4y - x + 2$

E2

Simplify:

$-x - 9 - x$

E3

Simplify:

$-4x + 4x + 8x$

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Simplify:

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ALG-3B24-L1 · E

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Algebraic Simplification

Like and Unlike Terms

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E1

Simplify:

$-x - 4y - x + 2$

E2

Simplify:

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E3

Simplify:

$-4x + 4x + 8x$

ALG-3B24-L1 · E

Show your working

Teacher answers & working

Algebraic Simplification · Like and Unlike Terms

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S1

Simplify:

$-x - 6x - 8x$

Answer:

$-15x$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-x - 6x - 8x = (-1 + -6 + -8)x$

$= -15x$

Step 2: All like terms in x combined by adding coefficients.

Step 3: Write each simplified group once in the final expression.

S2

Simplify:

$-3x - 3y + 2x - 8$

Answer:

$-x - 3y - 8$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-3x - 3y + 2x - 8 = (-3 + 2)x + -3y + -8$

$= -x - 3y - 8$

Step 2: x terms combined; unlike terms remain separate.

Step 3: Write each simplified group once in the final expression.

S3

Simplify:

$3x + 3 + 2x$

Answer:

$5x + 3$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$3x + 3 + 2x = (3 + 2)x + 3$

$= 5x + 3$

Step 2: Combined like x terms; constant remains.

Step 3: Write each simplified group once in the final expression.

Teacher answers & working

Algebraic Simplification · Like and Unlike Terms

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S4

Simplify:

$4x + 3x + 2x$

Answer:

$9x$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$4x + 3x + 2x = (4 + 3 + 2)x$

$= 9x$

Step 2: All like terms in x combined by adding coefficients.

Step 3: Write each simplified group once in the final expression.

E1

Simplify:

$-x - 4y - x + 2$

Answer:

$-2x - 4y + 2$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-x - 4y - x + 2 = (-1 + -1)x + -4y + 2$

$= -2x - 4y + 2$

Step 2: x terms combined; unlike terms remain separate.

Step 3: Write each simplified group once in the final expression.

E2

Simplify:

$-x - 9 - x$

Answer:

$-2x - 9$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-x - 9 - x = (-1 + -1)x + -9$

$= -2x - 9$

Step 2: Combined like x terms; constant remains.

Step 3: Write each simplified group once in the final expression.

Teacher answers & working

Algebraic Simplification · Like and Unlike Terms

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E3

Simplify:

$-4x + 4x + 8x$

Answer:

$8x$

Working:

Step 1: Identify the like terms. Like terms must have the same variable part and the same powers.

$-4x + 4x + 8x \;=\; (-4 + 4 + 8)x$

$=\; 8x$

Step 2: All like terms in x combined by adding coefficients.

Step 3: Write each simplified group once in the final expression.